

AI Agent for Autonomous Carrier Rerouting using LangGraph

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Assignment: AI Agent Research & Development

Abstract

This project presents a multi-agent AI workflow for autonomous carrier rerouting in logistics. The system analyzes shipment disruptions, retrieves alternative carriers, selects the most suitable carrier, and generates a final notification. The solution is implemented using Python, LangGraph, LangChain, and the Llama 3.3 70B model served through the Groq platform.

1. Introduction

Logistics companies frequently experience shipment delays because of weather, traffic, and operational issues. Manual rerouting is time-consuming and can delay deliveries. This project demonstrates how multiple AI agents coordinated by LangGraph can automate disruption analysis and routing decisions.

2. Problem Statement

The objective is to automate shipment rerouting by analyzing telemetry alerts, determining disruption severity, retrieving alternative carrier options, selecting the best carrier, and notifying the logistics team.

3. System Architecture

Workflow:

Telemetry → Triage Agent → Route Agent → Decision Agent → Notification → End.

The Triage Agent classifies shipment severity. The Route Agent retrieves carrier options. The Decision Agent selects the optimal carrier based on ETA and priority. The Notification Agent generates the final response.

4. Implementation

The application is implemented using Python. LangGraph orchestrates the workflow while LangChain connects the LLM. Groq provides inference for the Llama 3.3 model. Pydantic is used for structured outputs and python-dotenv manages API keys. A shared state allows agents to exchange information.

5. Results

The workflow executed successfully from telemetry input to notification output. Conditional routing was implemented, and the Decision Agent selected the most appropriate carrier with an explanation for the decision.

```
Telemetry
{'shipment_id': 'SHIP001', 'carrier': 'BlueDart', 'destination': 'Mumbai', 'delay_hours': 12, 'reason': 'Heavy rainfall'}

Triage
{'severity': 'High', 'action': 'reroute', 'priority': 'Immediate'}

Routes
[{'carrier': 'FedEx', 'eta_hours': 18, 'cost': 150}, {'carrier': 'DHL', 'eta_hours': 24, 'cost': 120}]

Decision
{'selected_carrier': 'FedEx', 'reason': 'Faster ETA and high priority shipment requires speed over cost consideration'}

===== NOTIFICATION =====

Shipment ID : SHIP001

Destination : Mumbai

Selected Carrier : FedEx

Reason :
Faster ETA and high priority shipment requires speed over cost consideration
```

6. Model Comparison

The workflow was successfully evaluated using Llama 3.3 through Groq. The architecture was designed to be model-independent using a shared `llm_factory` module. An attempt was made to integrate Google's Gemini model; however, compatibility issues between the integration library and the available Gemini model prevented execution within the project timeline. The comparison framework remains ready for future evaluation.

7. Challenges

Major challenges included prompt engineering for structured outputs, coordinating multiple agents through a shared state, and Gemini integration compatibility.

8. Future Work

Future improvements include integrating real logistics APIs, adding human approval for high-value shipments, benchmarking multiple LLMs, and deploying the application using FastAPI and Docker.

9. Conclusion

This project demonstrates a modular and explainable multi-agent AI system for logistics rerouting. LangGraph enables organized workflow execution, while Llama 3.3 provides intelligent decision-making. The architecture can easily be extended with additional models and real-world logistics services.

References

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3. <https://console.groq.com/docs>
4. <https://ai.google.dev/>
5. <https://docs.python.org/>